Data modeling: CSCI E-106

Applied Linear Statistical Models

Chapter 11 – Building the Regression Model III: Remedial Measures

Ridge Regression

- Ridge regression: a method of overcoming serious multicollinearity

problem by modifying the method of least squares.

- allowing biased estimators of the regression functions
- When an estimator has only a small bias and is substantially more precise than an unbiased estimator, it may well be the preferred estimator since it will have a larger probability of being close to the true parameter value.

Ridge Regression, cont'd

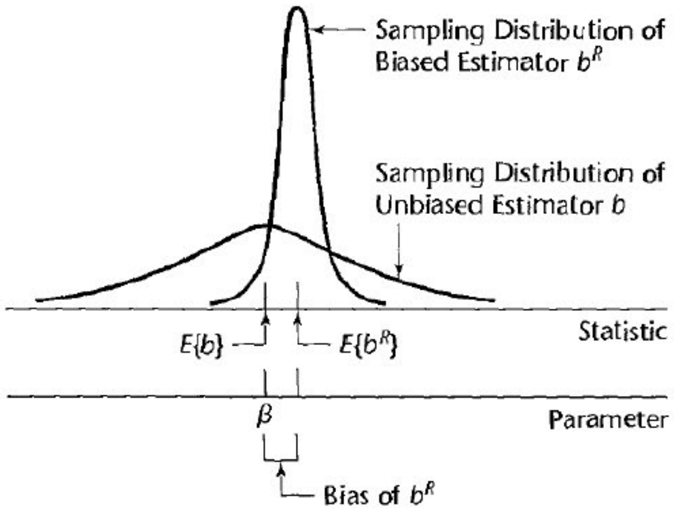

FIGURE 11.2 Biased Estimator with Small Variance May Be Preferable to Unbiased Estimator with Large Variance.

- Estimator b is unbiased but imprecise
- Estimator b^R is much more precise but has a small bias
  $\Rightarrow$ The probability that b^R falls near β is much greater than that for b

Ridge Regression, *cont'd*

- Ridge regression is very similar to least squares, except that the Ridge coefficients are estimated by minimizing a slightly different quantity. In particular, the Ridge regression coefficients β are the values that minimize the following quantity:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = \mathrm{RSS} + \lambda\sum_{j=1}^{p}\beta_j^2$$

- Here, $\lambda \geq 0$ is a tuning parameter, to be determined separately.
- The term $\lambda \sum_{j=1}^{p} \beta_j^2$ is a shrinkage penalty that decreases when the β parameters withdraw (shrink) towards the zero.
- Parameter λ controls the relative impact of the two components: RSS and the penalty term.
  - If $\lambda = 0$, the Ridge regression coincides with the least squares method.
  - If $\lambda \to \infty$, all estimated coefficients tend to zero.
- The regression Ridge produces different estimates for different values of λ. The optimal choice of λ is crucial and is usually done with cross-validation

Ridge Regression, *cont'd*

Ridge regression does have one obvious disadvantage. Unlike best subset, forward stepwise, and backward stepwise selection, which will generally select models that involve just a subset of the variables, ridge regression will include all p predictors in the final model.

The penalty $\lambda \sum_{j=1}^{p} \beta_j^2$ will shrink all of the coefficients towards zero, but it will not set any of them exactly to zero (unless$\lambda = \infty$)).

This may not be a problem for prediction accuracy, but it can create a challenge in model interpretation in settings in which the number of variables p is quite large.

Lasso Regression

- The *lasso* is a relatively recent alternative to ridge regression that over-comes this disadvantage. The lasso coefficients, $\hat{\beta}_\lambda^L$, minimize the quantity

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = \mathrm{RSS} + \lambda\sum_{j=1}^{p}|\beta_j|$$

- we see that the lasso and ridge regression have similar formulations. The only difference is that the penalty term, where ridge penalty term $\sum_{j=1}^{p}\beta_j^2$ is replaced by $\sum_{j=1}^{p}\left|\beta_j\right|$ in the lasso penalty
- Ridge uses L2-norm, whereas Lasso goes with L1-norm as penalty terms.

Lasso Regression, *cont'd*

- In Ridge regression, the penalty is the sum of the squares of the coefficients, and for Lasso, it's the sum of the absolute values of the coefficients. It's a shrinkage towards zero using an absolute value (L1-norm) rather than a sum of squares (L2-norm).
- These are two normalization procedures (Loss functions) commonly used in the machine learning algorithm.
  - L1-norm is also known as least absolute deviations, or least absolute errors. It is basically minimizing the sum of the absolute differences between the target value and the estimated values.
  - L2-norm is also known as least squares. It is basically minimizing the sum of the square of the differences between the target value and the estimated values.
- The Ridge regression produces a model with all the variables, of which the part with coefficients is closer to zero. Increasing λ forces more coefficients to be close to zero, but almost never exactly equal to zero, unless $\lambda = \infty$. For forecasting this is not a problem, while interpretation can sometimes be problematic.
- Lasso regression tries to overcome this aspect. The Lasso regression penalty term, using the absolute value (rather than the square, as in the regression Ridge), forces some coefficients to be exactly equal to zero, if λ is large enough. In practice, Lasso automatically performs a real selection of variables.

Elastic Net

- ElasticNet is a hybrid of both Lasso and Ridge regression. It is trained with both L1-norm and L2-norm prior as regularizer. A practical advantage of trading off between Lasso and Ridge is that it allows ElasticNet to inherit some of Ridge's stability under rotation.

$$\frac{\sum_{i=1}^{n}(y_i - x_i^J \hat{\beta})^2}{2n} + \lambda \left(\frac{1-\alpha}{2} \sum_{j=1}^{m} \hat{\beta}_j^2 + \alpha \sum_{j=1}^{m} |\hat{\beta}_j| \right)$$

Elastic net regularization

- In addition to setting and choosing a lambda value elastic net also allows us to tune the alpha parameter where $\boldsymbol{\alpha}$ = 0 corresponds to ridge and $\boldsymbol{\alpha}$ = 1 to lasso.
- Simply put, if you plug in 0 for alpha, the penalty function reduces to the L1 (ridge) term and if we set alpha to 1 we get the L2 (lasso) term. Therefore we can choose an alpha value between 0 and 1 to optimize the elastic net.
- Effectively this will shrink some coefficients and set some to 0 for sparse selection.